Exp:01

sbai26

Exp:01 Exploring Matplotlib, Seaborn & Ploty

Aim:-

Take a dataset and perform visualization using matplotlib, Seaborn & ploty.

Libraries used :-

* Pandas
* Numpy
* Matplotlib
* Seaborn
* Ploty

Procedure :-

* Load the dataset into Python
* Import the required libraries
* Check for missing values in the dataset.
* List column names
* Drop rows or column having missing values.
* Remove any feature consider unimportant.
* Create Bar plot, Scatter plot, Box plot, Pie Chart, Histogram.
* Interpret the insights from following visualization.

Code :-

```
# Import libraries:-
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

Load the dataset

```
df = pd.read_csv("Titanic-Dataset.csv")
df.isnull().sum()
df = df.dropna(subset=["Embarked"])
```

Histogram Box-Plot:

```
import
plt.figure(figsize=(8,5))
sns.countplot(x="Parch", hue="Survived", data=df)
plt.title("Survival Count by Parch")
plt.xlabel("No. of Parent/Children")
plt.ylabel("No. of Passengers")
plt.legend(title="Survived", labels=["Died","Survived"])
```

3) Pie-Chart:

```
import matplotlib as plt
Southampton = df[df["Embarked"]=="S"]["Age_Gro
    .value_counts()
plt.figure(figsize=(6,6))
plt.plot(
        S.outhampton,
        labels = southampton.index,
        autopct = "%1.1f%%",
        startangle = 90
)
plt.title("Age Group of Passengers - Southampton (S)")
plt.show()
```

Histogram:

```
plt.figure(figsize=(8,5))
sns.histplot(data=df,
             x="Age",
             hue="Pclass",
             bins=20,
             multiple="Stack"
)
plt.title("Age Distribution by Passenger Class")
plt.xlable("Age")
plt.ylabel("No: of Passengers")
plt.show()
```

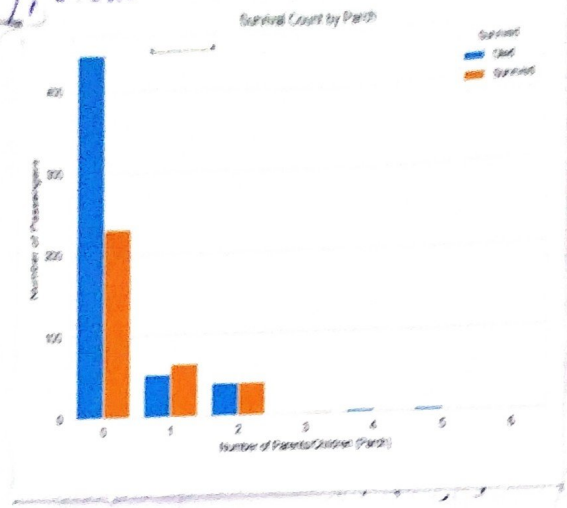
Box Plot:

```
import matplotlib as plt
import seaborn as sns
plt.figure(figsize=(8,5))
sns.boxplot(x="Pclass", y="Fare", data=df)
plt.title("Ticket Fare Distribution by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Ticket Fare")
plt.show()
```

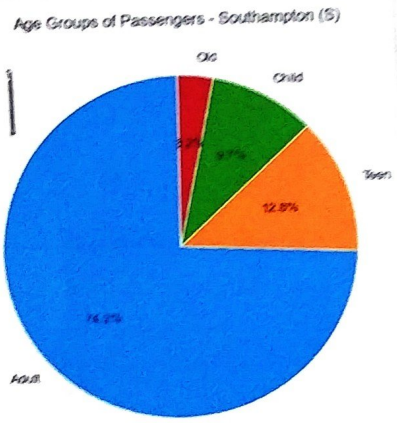
Scatter Plot

```
import seaborn as sns
import matplotlib.pyplot as plt.
plt.figure(figure=(8,5))
sns.histplot(df["Fare"], bins=20, kde=true)
plt.title("Distribution of Ticket Fare")
plt.xlabel("fare")
plt.ylabel("No: of Passengers")
plt.show()
```

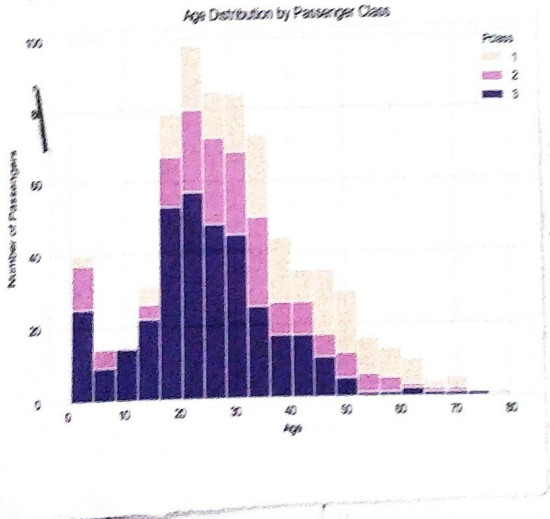
Visualization



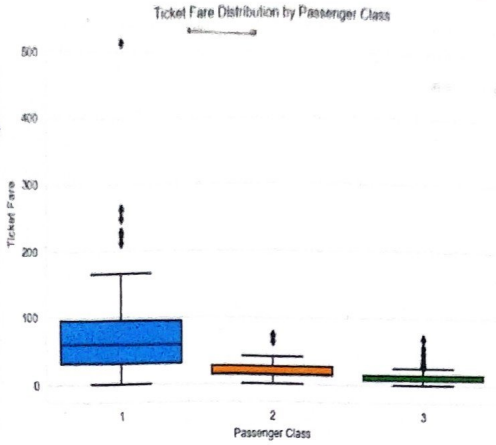
* It shows the survival count count for Parent with children.



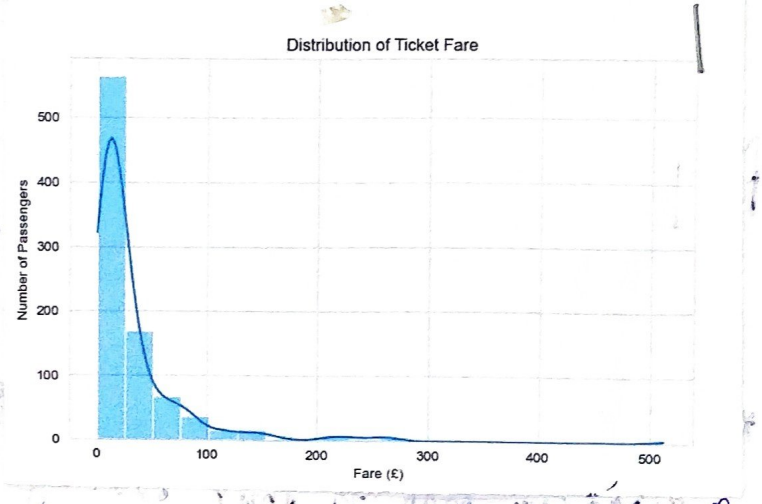
* It shows from Southampton which age groups are taken in the Titanic



* It shows which age people are in which P-class.



It shows for each P-class how much range of ticket amount is collected.



* It shows the ticket amount given by no: of Passengers.

Result:- 22/7/24

By taking a dataset exploration of Matplotlib, Seaborn and plotly visualization is done & executed.